The frequency of the oscillation is

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{k}{m}} \text{ Hertz} \qquad (10.23)$$

and the time period of the oscillation is

$$T = \frac{1}{f} = 2\pi\sqrt{\frac{m}{k}} \text{ seconds} \qquad (10.24)$$

Notice that ***in simple harmonic motion, the time period of oscillation is independent of amplitude***. This is valid only if the amplitude of oscillation is small.

The solution of the differential equation of a SHM may be written as

$$x(t) = A\sin(\omega t + \varphi) \qquad (10.25)$$

Or

$$x(t) = A\cos(\omega t + \varphi) \qquad (10.26)$$

where A, ω and ϕ are constants. General solution for differential equation 10.21 is $x(t) = A\sin(\omega t + \varphi) + B\cos(\omega t + \varphi)$ where A and B are contants.

Note

(a) Since, mass is inertial property and spring constant is an elastic property,

Time period is $T = 2\pi\sqrt{\frac{m}{k}}$

$$T = 2\pi\sqrt{\frac{\text{Inertial property}}{\text{Elasticproperty}}} = 2\pi\sqrt{\frac{\text{displacement}}{\text{acceleration}}}$$

(b) $\frac{\text{Displacement}}{\text{acceleration}} = \frac{x}{\frac{d^2x}{dt^2}} = -\frac{m}{k}$, whose modulus value or magnitude is $\frac{m}{k}$

hence, time period $T = 2\pi\sqrt{\frac{m}{k}}$

10.4.2 Vertical oscillations of a spring



Figure 10.14 Springs

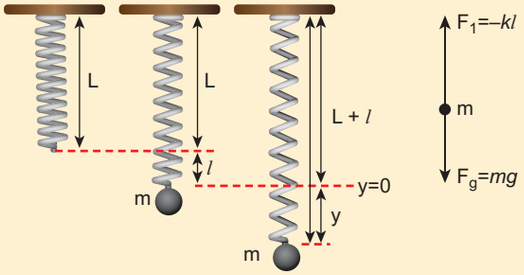


Figure 10.15 A massless spring with stiffness constant k

Let us consider a massless spring with stiffness constant or force constant k attached to a ceiling as shown in Figure 10.15. Let the length of the spring before loading mass m be L. If the block of mass m is attached to the other end of spring, then the spring elongates by a length l. Let F_1 be the restoring force due to stretching of spring. Due to mass m, the gravitational force acts vertically downward. We can draw free-body diagram for this system as shown in Figure 10.15. When the system is under equilibrium,

$$F_1 + mg = 0 \qquad (10.27)$$

But the spring elongates by small displacement l, therefore,

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